Environment and Preferences

A two-period OLG economy: Y_t young and O_t old share a fixed housing stock $\bar{H}$.

- Entrants draw income and liquid wealth, (y_i, b_i), from a fixed distribution F.
- Young choose consumption, housing, fertility, saving, and tenure.
- Old choose consumption, housing, and an estate.

Flow utilities are

$$u_t^Y(c, h, n) = \log(c - \chi n) + \alpha \log(h - \kappa n) + \vartheta_t \log n,$$

$$u^2(c^2, h^2, e) = \log c^2 + \gamma \log h^2 + \omega_B \log e.$$

Each child needs χ goods and κ housing; e gives warm-glow utility.

Old-age utility is discounted by β; owning adds the taste ξ_i.

Housing and Finance

A mortgage finances share ϕ_t of a home bought at price P_t:

$$\underbrace{h \leq h_R^{\max} < h_O^{\max}}_{\text{rent}} \qquad \underbrace{h \leq h_O^{\max}, \quad (1-\phi_t)P_t h \leq b_i}_{\text{own}}.$$

A buyer may afford the cost of using a home but lack its down payment.

A world bond has price q and gross return $R_f = q^{-1}$. With next-period rent r_t and property-tax rate τ^p, rental entry implies

$$qr_t = (1 + q\tau^p)P_t - qP_{t+1} > 0.$$

- Income and rebates arrive after closing; unsecured borrowing is unavailable.
- Old owners may downsize, but cannot buy more or let retained housing.

Competitive Equilibrium

Households optimize given the price path; initial cohorts and old assets are fixed.

Housing clears; property-tax revenue is rebated equally as T_t:

$$Y_t \bar{h}_t^Y + O_t \bar{h}_t^O = \bar{H}, \qquad (Y_t + O_t) T_t = q\tau^p P_t \bar{H}.$$

Here $\bar{h}_t^Y, \bar{h}_t^O$ are average housing by age, across both tenures.

Mean completed fertility $\bar{n}_t$ determines the next young cohort:

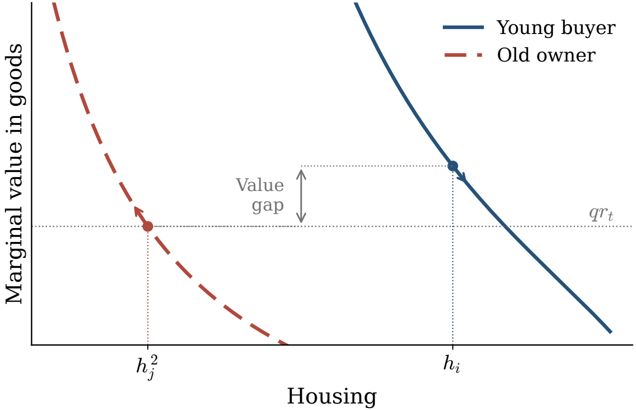
$$Y_{t+1} = \nu \bar{n}_t Y_t, \qquad O_{t+1} = Y_t,$$

where ν converts children into entering households.

A positive steady state has constant prices and allocations, $Y^* = O^*$, and $\bar{n}^* = 1/\nu$.

Housing Misallocation

Marginal housing value measures willingness to pay for extra space.



$$MV_i^Y = \frac{\alpha(c_i - \chi n_i)}{h_i - \kappa n_i}$$

$$MV_j^O = \frac{\gamma c_j^2}{h_j^2}$$

$$MV_i^Y > qr_t = MV_j^O.$$

Young down payments bind; saving is positive; owner size, retention, and estate limits are slack.

A Compensated Reallocation

Move a little housing from old owners to young owners with a higher marginal value.

- Compensate the old with goods from the young.
- Keep fertility, tenure, estates, and future real allocations fixed.
- Respect physical limits and total resources; relax private financing limits.

Proposition. The equilibrium is inefficient at fixed fertility: some young households can gain while everyone else is equally well off.

The value gap holds on positive masses of feasible pairs. There is no real moving cost, and existing financial claims are honored.

Housing and Fertility

Relax a young buyer's binding housing limit. Allow fertility to adjust, holding prices and tenure fixed.

Fertility rises if and only if the space benefit exceeds the resource cost:

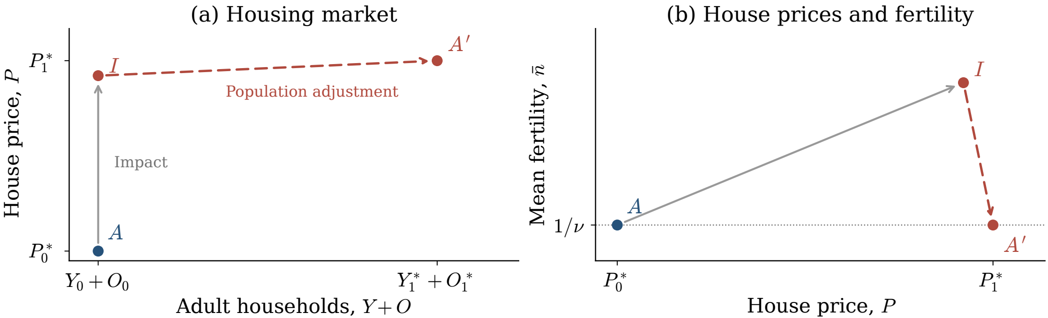
$$\underbrace{\frac{\alpha\kappa}{(h-\kappa n)^2}}_{\text{space benefit}} > \underbrace{\frac{\chi q r_t}{[1+\beta(1+\gamma+\omega_B)](c-\chi n)^2}}_{\text{resource cost}}.$$

More space helps accommodate children; paying for it uses other resources.

Positive saving and interior old-age choices; the same constraints bind and no transfers accompany the change.

Equilibrium Transition

After a small permanent credit relaxation, $\phi_1 > \phi_0$:



A: initial steady state; I: impact; A': new steady state.

Local example with both tenures: fertility rises initially, then returns to replacement with more households. Arrows compare equilibria; intermediate adjustment can overshoot.